

Research on the Application of Low-Cost Aerial-Ground Delivery System Using UAV-UGV Joint

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Abstract—The delivery service using unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) collaboration has been a hot topic in the industry, and the low-cost UAV-UGV aerial-ground delivery system research has significant practical significance. This study proposes a low-cost aerial-ground delivery system for residential areas, based on UAV-UGV collaboration. The system uses a 2D SLAM enabled UGV for precise ground transportation and a quadrotor UAV equipped with SVO for high-altitude delivery of small and medium-sized packages. Both the UGV and the UAV are developed based on the Robot Operating System (ROS). This study comparing energy consumption and time taken under different operating modes and visual odometers respectively.. The delivery process is divided into three stages and a visual-maximization-based SVO fusion method is proposed for the UAV search in case of target loss. Experimental results show that the proposed system is effective, accurate, timely, and has high practical value.

Keywords—UAVs, UGVs, aerial-ground delivery system, SVO

I. INTRODUCTION

The rapid growth of logistics demand and consumer demand for faster delivery speeds has led to the proposal of using UAVs for goods delivery by companies such as Amazon, Google, Meituan, and JD.com [1]. Successful cases have shown that small parcels can be directly delivered to rural area consumers' balconies using GPS and the UAV's height limitlessness [2]. Hence, efficient and low-cost UAV delivery systems have important practical significance and application prospects.

Although Unmanned aerial vehicles (UAVs) have shown good performance in many fields, their use of GPS to obtain latitude and longitude information is prone to signal failure due to the dense construction of buildings. Furthermore, due to their weight limitations, using UGVs for delivery over the same distance is the lowest cost option. Therefore, researching the coordinated delivery of UAVs and UGVs is of great significance [3]. As research into unmanned aerial vehicle-unmanned ground vehicle systems provides support for our work [4].

Despite an increasing number of joint delivery schemes using UAVs and UGVs, research on cost minimization remains limited. To achieve a minimum cost

delivery plan, it is necessary to reconsider the problem from the perspective of UAVs, UGVs, and their coordinated cooperation. Safety is a critical issue in UAV delivery, requiring the development of predictable collision avoidance algorithms as described in literature [5]. Due to GPS signal instability, visual SLAM technology can improve UAV positioning accuracy, as discussed in literature. Visual SLAM technology is increasingly used in UAVs, as surveyed in literature [6][7].

Z. Chao et al. present offline monocular visual odometry (VO) for robots, which is significant for our experiments [8][9]. We use the VO part of visual SLAM (vSLAM) as the local trajectory planning scheme for our UAV. Additionally, O. Omotuyi and M. Kumar optimize UAVs by coupling known thrust and dynamics with known control inputs and surrounding UAV dynamics and visual inertial measurements using the Unscented Kalman Filter (UKF) [10], providing theoretical support and experimental guidance for our low-cost, high-efficiency UAV-UGV joint delivery system development.

Considering the generally flat ground in residential environments, multi-line LiDAR is more expensive and computationally demanding than single-line LiDAR. Shen et al. conducted an in-depth analysis and comparative study of LiDAR SLAM algorithms for mobile robots [11], finding that the Cartographer scheme has stronger real-time performance in low-cost environments.

R. Xuan et al. summarized five forms of coordination for delivery: joint, parallel, UGV/UAV-supported, and mixed delivery [12]. The topology of the problem can be abstracted into a TSP model for research. M. Shan et al. developed a two-stage solution method to optimize total cost for truck-UAV logistics based on a "maximum weight judgment method" and improved simulated annealing algorithm [13]. For multiple loop deliveries, A. M. El-Adle et al. proposed a Variable Neighborhood Search (VNS) heuristic algorithm to solve the TSP-D and a two-stage strengthening scheme used by VNS [14].

Although proposed solutions have achieved good performance in specific scenarios, they often require more cost and computing resources and are limited to general environments. Our research aims to achieve cost-effective autonomous UAV-UGV joint delivery in fixed areas such as

dense residential areas. Our goal is to minimize costs while maintaining real-time performance. This paper proposes a UAV-UGV joint autonomous delivery scheme and conducts experiments and analysis to provide theoretical support and practical guidance.

II. SYSTEM FRAMEWORK

A. Hardware Components

In order to explore more cost-effective and faster delivery solutions, we initially propose a combination of unmanned aerial vehicles (UAVs) and utility ground vehicles (UGVs), and then compare it with other options. To achieve cost minimization, our UAV hardware configuration includes an Inertial Measurement Unit (IMU) for controlling the UAV's attitude, a main control board and motion control board, a quadrotor UAV, and a forward-facing monocular camera (used for target search). We also use a wifi module to communicate with the UGV, connecting to wifi through a self-built hotspot for communication.

The UAV hardware includes an IMU for attitude control, main and motion control boards for flight control and navigation, and a monocular camera for visual perception and recognition. The UGV hardware comprises a main control board, single-line lidar for high-precision perception and modeling, and a Mecanum wheel vehicle with an Ackermann structure for good maneuverability and versatility in indoor and outdoor environments.

B. Submodules and Functions

Fig.1 shows the hardware configuration and corresponding functionality of the entire system.

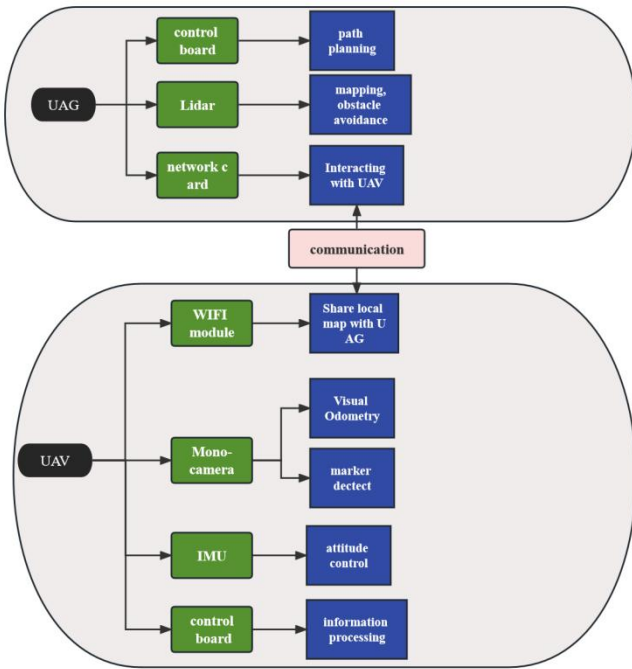


Fig. 1. Functional description of each module of UGV-UAV system

The black rectangular block represents the system divided into two loosely coupled subsystems: the UGV part and the

UAV part. The green rectangular block represents each hardware module, while the blue rectangular block represents the corresponding role and function of the hardware module in the system. Communication plays a crucial role in the system, as indicated by the pink rectangular block. This module includes the function of the UGV sending the start transport command to the UAV and the function of map sharing between the UAV and UGV.

In the UGV part, we use a Mecanum wheel vehicle with an Ackermann structure chassis and a single-line lidar for environment perception and modeling. The UGV's main control board employs an Intel I5 processor for system efficiency and stability. For cost-effectiveness, we selected the EAI G1 high-precision lidar for the single-line lidar hardware device. During delivery, the ROS system is used for autonomous UGV navigation, enabling precise positioning and path planning. To ensure efficient coordination between the UGV and UAV, we equip a wireless network card for bidirectional communication between the two systems.

In the UAV part, we utilize a forward-facing monocular RGB camera as the sole observation sensor. Its main function is to detect the delivery area location and establish a local map during ascent and return phases. Using SVO visual odometry technology, the camera also estimates depth via triangulation, enabling obstacle avoidance. A Wi-Fi module facilitates information exchange and command reception with the UGV, and the relative pose relationship between the UAV and UGV is transmitted through this module, ensuring a smooth UGV return.

The IMU of the UAV is a necessary module that can achieve attitude maintenance and stable flight control. The main control board uses the Intel Aero Compute Board, running the ROS system. We designed multiple state transitions, including standby, ascent of delivery packages, and return, to achieve autonomous control and navigation of the UAV in different stages. Through the above hardware configuration and functionality, efficient delivery and autonomous navigation of the UAV can be achieved.

It is worth noting that the driving control boards of both the UAV and UGV utilize STM32 chips, which play a critical role in receiving motion control commands from their respective main controls and converting them into motor drive power through the drive module. While they are not depicted in the figure due to their weak relationship with the system design, their existence is indispensable for the proper functioning of the entire system. Although the communication and information processing between the two systems are functionally performed by their respective main controls, it is facilitated through a hardware wireless network card supporting AP mode. An AP hotspot is established on the UGV, and the two systems exchange information via the Wi-Fi protocol, enabling efficient coordination and information sharing.

C. Execution Logic

In the context of a residential area, our system's operation process is shown in Fig. 2 and consists of two stages. The first stage involves UAG autonomous navigation mapping, with

only one mapping task required for a fixed area system, allowing all systems to share the same map. When delivery is needed, the UGV carrying the UAV autonomously navigates to the destination and instructs the UAV to transport the parcels. The UAV utilizes visual odometry for transportation and the coordinate transformation relationship between the UGV and UAV to minimize the UAV's path. Following delivery, the UAV returns to the UGV for the next task or recharge. Throughout the process, the UAG system efficiently coordinates the UGV and UAV, ensuring efficient delivery services.

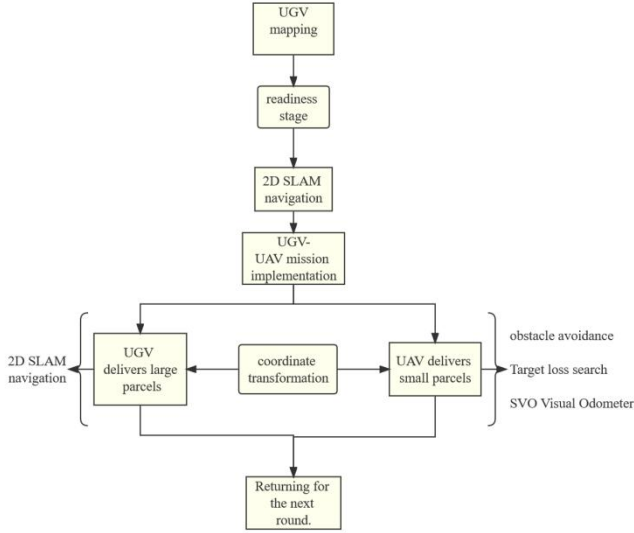


Fig. 2. UAV-UGV Aerial-Ground distribution system execution logic

For packages that are heavy for delivery by UAV, the system adopts a method of notifying the recipient in advance to pick up the package at a designated location. During the UAV delivers small packages, the UGV will deliver the large packages to the designated location, which increases the difficulty of the UAV returning to the UGV. Fortunately, our UAV uses an application that relies on visual odometry (SVO) and can construct maps of the surrounding environment during the end delivery process. In addition, the UAV and UGV need to communicate over short distances to achieve map sharing based on the UAV's local map. Through coordinate transformation, we can build the coordinate transformation relationship between the UAV and the UGV, thereby ensuring that the UAV can successfully "return".

In our system, we use SVO for high-precision UAV positioning and map construction. Coordinate transformation technology enables effective communication and coordination between the UAV and UGV, improving delivery efficiency and accuracy. If the recipient marker is not detected during delivery, the UAV performs a target search based on the maximum parallel view of the plane and returns home if unsuccessful. We utilize visual sensors and deep learning for obstacle avoidance functionality during UAV movement. Additionally, a maximum parallel view of the plane algorithm is used for target search and positioning.

III. IMPLEMENTATION

A. Delivery Area Detection

When selecting a delivery area detection method, we prioritized low cost and high detection speed. We evaluated various mainstream target detection algorithms, including traditional algorithms such as SIFT, SURF, and deep learning methods such as RCNN, SSD, and YOLO. After conducting experiments and comparisons, we found that YOLO, SSD, and traditional target detection algorithms have better stability and real-time performance in solving the problem of instance detection. However, deep learning-based algorithms require significant upfront preparation and consume computational resources, and the characteristics of different household types in residential areas are inconsistent, increasing the difficulty of detection tasks. In comparison, traditional visual methods utilizing templates for target detection also demonstrate remarkable performance. Ultimately, we chose template-based traditional target detection methods to complete this task.

We selected the WhyCon template from the optional visual markers because of its seamless integration with the ROS system, fast running speed, and long detection distance. Experimental results verified the correctness of our choice, as it reduced costs without compromising running speed. By selecting appropriate algorithms and templates, we efficiently implemented target detection and tracking, enabling autonomous navigation and positioning of the system. Our choices also considered the balance between cost and performance, improving the system's scalability and practicality.

B. 2D Navigation Algorithms

Accurate map construction is a crucial step in the field of SLAM, as it ensures navigation accuracy. When selecting a 2D SLAM algorithm for our system, we carefully weighed the advantages and disadvantages of various algorithms, given our use of a single-line lidar. F. Lu et al. [15] have presented several feasible solutions, including Gmapping, Cartographer, Hector, and others, each with unique strengths and weaknesses. Ultimately, we selected the Cartographer algorithm for our system.

Our selection of a SLAM algorithm was based on considerations of applicable scenarios and hardware requirements. GMAPPING lacks loop closure detection and is not suitable for constructing large scenes, while Hector has high requirements for laser radar and does not utilize odometry information. Karto is suitable for large-scale environments, but Cartographer, which utilizes odometry and IMU, has global and local maps, backend optimization, and low cumulative error, even with low-cost laser radar. Therefore, we chose Cartographer for map building, enabling efficient map construction and autonomous navigation. This selection ultimately supports the successful realization of autonomous control and positioning for our UAV-UAG joint delivery system.

C. Visual Odometry

Visual localization and local map establishment are critical functions in our system, enabling autonomous control and

positioning of UAVs. For our single-camera equipped robot, we carefully selected a suitable visual localization algorithm from the many excellent algorithms available in the field of vSLAM, including ORB, ROVIO, and SVO. ORB is a feature-based algorithm with excellent performance in the VSLAM field, while ROVIO is a low computational cost, image block-based filtering VIO algorithm, though residual error issues exist. SVO combines the advantages of feature-based and direct methods and uses parallel tracking and mapping approaches to quickly and accurately localize the robot. Based on our experiments, we selected SVO as the visual odometry algorithm for our robot.

D. Path to Maximize View

During UAV delivery, errors in recipient, UAV, or UGV may cause the UAV to lose its target, and the ascent path may have no visual markers detected. To address this, we propose a path minimization method that utilizes robot path planning and target search to optimize the robot's path and maximize the search field of view, improving target search efficiency.

When the UAV has not found the target along the ascent path, it retreats to the maximum detection distance of the visual marker (7 meters) to obtain the maximum field of view, as shown in Fig. 3.

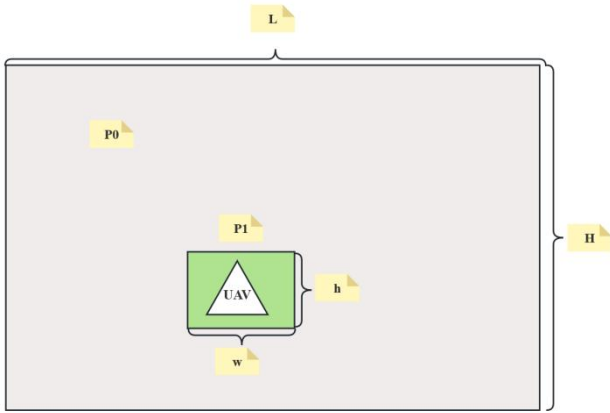


Fig. 3. Schematic of a drone flight when the target is lost

Assuming the height of the building is H and the length is L , and the visible field of view of the UAV at this time is denoted as w and h . The camera's intrinsic and extrinsic parameters and transformation relationship are represented by equation (1):

$$\begin{cases} u = \frac{f_x}{z} x + c_x \\ v = \frac{f_y}{z} y + c_y \end{cases} \quad (1)$$

To analyze this problem, we can obtain the relationship between u and v as shown in equation (2) since our image format is 640x480.

$$u/v = 640/480 \quad (2)$$

To simplify the problem, we consider that the parameters f_x , f_y , c_x , c_y in the camera's intrinsic matrix are usually small, so they can be approximated as equal. Based on this, we assume that the true target point is located at p_0 , and the position of the UAV is p_1 . The error e should follow a Gaussian distribution as shown in equation (3), where $u = 0$.

$$e = |p_1 - p_0| \sim N(\mu, \delta^2) \quad (3)$$

To minimize the path and expand the search field of view, we propose a robot path planning and Gaussian distribution-based target search algorithm. A mathematical model is constructed, as shown in Fig.4, with the center point determined by solving equation (4). We search within the range of the distribution model's error on the left and right sides to expand the search field of view around the center point of the Gaussian distribution. The search field of view is rectangular with length and width of , and we use a "surrounding" search strategy. By gradually expanding the search range with a step length of , the target can typically be found within the range of .

$$\arg \min_{p_1} (e) = \arg \min (|p_1 - p_0|) \quad (4)$$

If the object is beyond this range, we no longer consider the case that conforms to this distribution. We prioritize the shortest path and continue to search quickly with a step length of $2w$ on one side before switching to the other side, and finally returning the same way.

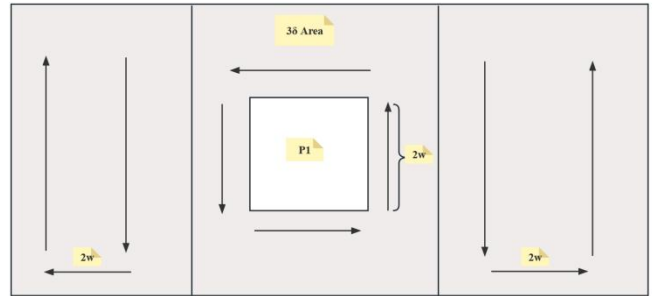


Fig. 4. Schematic of UAV flight based on Gaussian distribution and path-minimum field of view maximization after loss of target

IV. EXPERIMENTAL RESULTS

A. Simulation

The cost of on-site experiments is relatively high. In previous experiments, scientists usually conducted on-site experiments only after obtaining good results in simulations. Therefore, we first simulated the problem in the Gazebo world and kept its structure configuration consistent with the final on-site configuration. Fig.5 shows the UAG-UAV platform navigating to the ground floor.

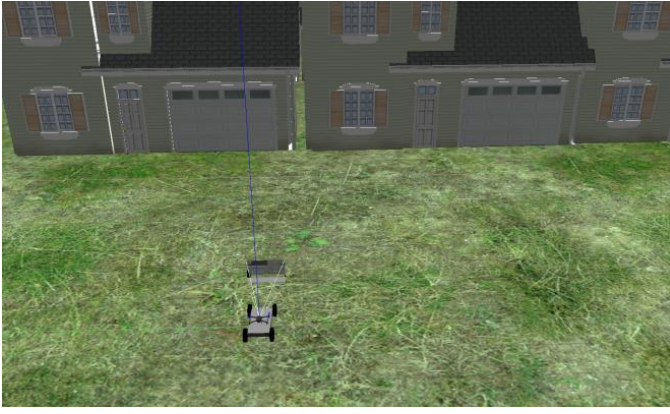


Fig. 5. Stage one: The Preparation of Systems

Fig.6 shows Unmanned ground vehicles (UGVs) and UAVs are currently executing distinct missions.



Fig. 6. Stage two : UGVs and UAVs are executing missions

Fig.7 shows the coordinate relationship obtained using SVO during the parallel working phase and displayed in RVIZ, including a small piece of camera output. It can be seen from the figure that the correct recognition position is clearly found.

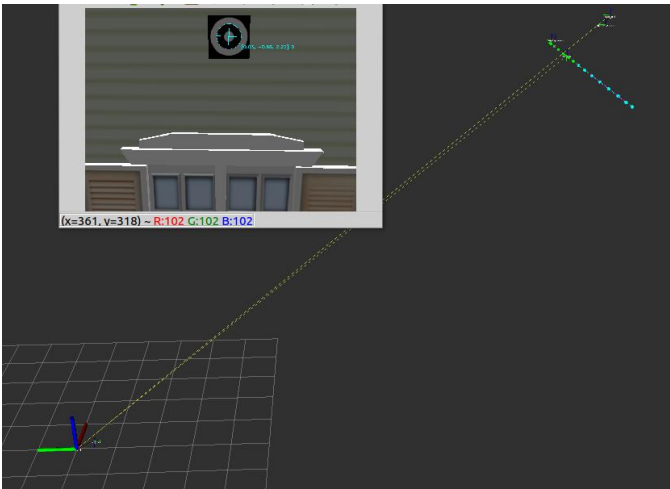


Fig. 7. the coordinate relationship in stage two

Fig.8 represents the completion of one round of tasks, and the UAV returns successfully.



Fig. 8. Stage three: The UAV successfully returned to the UGV

B. Field Test

The UAV and UGV have five different operation modes, and we conducted experimental analyses on the energy consumption for each mode. The UGV chassis weighs about 10 kg, and the battery is a power lithium battery with a capacity of 11200 mAh/24V. The UAV weighs about 1.3 kg, and the battery is a 3S model aircraft battery with a capacity of 5200 mAh. The experiments were conducted after the mapping phase was completed, and the goal of one round of tasks was to successfully deliver the parcel and return. Table I shows that the ground-air parallel operation mode has the lowest energy consumption, which is why we chose this mode as the overall mode for our low-altitude delivery system.

TABLE I. ENERGY COSTS

<i>Delivery Modes</i>	<i>Energy(mAH)</i>
Collaborative Delivery	3472.3
Parallel Delivery	4226.6
UGV-Assisted UAV	3576.7
UAV-Assisted UGV	5093.3

For real-time considerations, we chose three types of visual odometry methods: SVO, ORB, and ROVIO. To demonstrate the average processing time of these three methods in our system, we conducted experiments with each of them applied to our system (under monocular camera conditions). The experimental results are shown in Table II.

TABLE II. COMPARISON OF DIFFERENT VISUAL ODOMETRY

<i>Types of VO</i>	<i>Average Processing Times(ms)</i>	<i>Trajectory Error(m)</i>
SVO	17.7	0.10
ORB	28.1	0.22
ROVIO	34.3	Discarded

It can be seen that the processing time of the three algorithms is between 20-35ms, which meets the requirements

of real-time operation. However, the processing time difference between ROVIO and SVO is nearly half, so we discarded ROVIO in this stage. Next, we compared the average trajectory error between ORB and SVO in our environment. It is obvious that SVO has a better performance, and its average trajectory error is an amazing 0.06 meters, compared to ORB. The analysis shows that this may be due to the experimental environment, and SVO performs better when the photometry is consistent. We chose SVO as our visual odometry method due to its excellent performance.

Fig.9 shows the UAV-UGV delivery system, with the UGV on the ground going to the first-floor collection point to place the parcels, while the UAV is approaching its target receiving location for cargo transportation. Our solution which adopts the mode of UAV and UGV working together and respectively running their respective functions.



Fig. 9. Field Test

V. CONCLUSIONS AND FUTURE OUTLOOK

Although more and more UAV delivery systems are limited by regional restrictions, human factors, no-fly zones, and other reasons, our work partially solves the limitations of UAV operation due to region and no-fly zone issues. To make it better and faster, we conducted research on this issue from the perspective of scenarios in residential areas and minimum energy consumption. Our UAV-UGV delivery system seems very complex, but the algorithms used in it are very simple and efficient. We believe that it can inspire more researchers to improve and perfect this topic in other scenarios.

This is a preliminary model, and there are still many areas that need to be improved and solved. For example, in actual delivery, there is more than one combination of UAV-UGV delivery systems, and it is important and inevitable to study more intelligent agent coordination processes between systems. In addition, when the number of UAV-UGV delivery systems increases, the route planning of UAV in the air also needs to be considered. One UAV responsible for one building is not

enough. This not only involves coordination between systems, but also involves the coordination of UAV path planning in the air and even re-planning the path of multiple UGVs on the ground. Finally, we need to equip the UAV with a gripper that can grab small packages. The future of low-altitude delivery of goods will definitely move towards cost reduction, and the solutions adopted in different fields will certainly be different. This technology can be realized through robot arm control and mechanical modeling technology, enabling the UAV to independently grab and deliver small packages. The application of this technology will greatly improve the delivery capacity and efficiency of UAVs, providing more possibilities for future UAV delivery services.

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